

Comparing Path Length by Boundary Following Fast Matching Method and Bug Algorithms for Path Planning

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Abstract. Local or sensor-based path planning in an unknown environment is a challenging problem for mobile robots. In completely known grid environments, the shortest Euclidean-length path connecting a given start-goal point pair can be generated using the Fast Marching Method (FMM). In contrast, in unknown environments, path planning must be reactive (i.e. based on recent sensory information) and *boundary following* (BF) is one type of reactive path planning technique. In [1], a hybrid method called Boundary Following FMM (BFFMM) was proposed that could extend the applicability of FMM to work well in an unknown environment using only local sensory perceptions. Bug algorithms are one family of navigation algorithms that also utilize BF technique in planning paths in unknown environments. This paper compares the path length of the paths generated by BFFMM with those by Bug1 and Bug2, the earliest versions of the Bug family algorithms to quantitatively reveal the effects of environment uncertainties on length of paths generated by navigation methods using boundary following. By comparison, BFFMM is a modest mobile robot path planner that bridges the gap between unknown and known terrain.

1 Introduction

Depending on whether the prior knowledge of the whole environment is available, the problems of path planning may be divided into two classes: local or sensor-based and global. For global path planning, or similar path planning with complete information, it is assumed that the mobile robot knows a priori the environment,

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and the main concern of path planning is efficiently generating a collision-free and/or optimal path. Many different methods of navigating mobile robots achieving varying degrees of success in a variety of conditions/criteria of motion and environments have been developed [9], [6]. In contrast, for local path planning, or similar path planning with uncertainty, the mobile robot has no a priori information about the environment in which it is moving around—an environment likely populated by obstacles whose positions, shapes, and quantities are unknown. The mobile robot locally updates its knowledge of environment during runtime by exploring using local information about proximate surroundings provided by its on-board sensors, allowing for planning a local path. The greatest concern when navigating an unknown path is generating a usable path to the destination.

For global path planning, the fast marching method (FMM) proposed by Sethian [7] has received more attention and has been applied to mobile robot path planning in a grid environment. Through wave propagation to solve the Eikonal equation, FMM can generate paths that closely resemble the geodesic via use of wave propagation to solve the Eikonal equation. Although FMM is a powerful tool for solving the problem of geodesic, shortest path planning, FMM is still limited to completely known terrain. In [1] we extended the applicability of FMM assuming the capability of boundary following (BF) by mobile robots and perfect sensing ability by combining it with the BF and developed a hybrid approach called Boundary Following FMM (BFFMM), which could be applied well to an unknown terrain. Bug Algorithms, are famous local path planning algorithms in an unknown planar environment utilizing boundary following and were invented by Lumelsky [3] and later a variety of Bug-like algorithms were proposed [3]-[5], [8]. A report was released recently that compared the performance of various Bug algorithms [4]. However, comparisons with other BF based path planning algorithms have not been made, and their optimality aspects have not been addressed. In this paper, we will compare the path length of paths generated by the BFFMM algorithm and the Bug algorithms, specifically the Bug1 and Bug2.

The paper is organized as follows. Section 2 introduces the technique used for experimental comparisons of the algorithms. Section 3 presents the simulation results in different environments and also discusses the difference between the different algorithms. The final sections include the conclusion and a possible future direction of research.

2 Local Navigation Methods Based on Boundary Following

In this paper, we make the following assumptions about the robot to which the simulations apply: The robot moves in an unknown 2D polygonal environment that is arbitrarily populated with a set of static polygonal obstacles. The robot is modeled as a point by enlarging the obstacles to account for the robot size. Moreover, the robot is equipped with a sensory system capable of exploring the environment by measuring the distance and direction to an obstacle within the visibility range of its sensors.

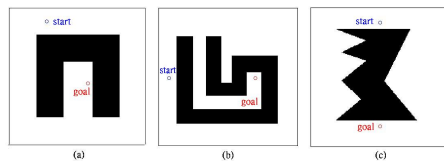
2.1 Boundary Following FMM (BFFMM) [1]

The BFFMM may be divided into an exploring stage and a path planning stage. In the exploring stage, the aim of the robot is to build its own partial map of the entire terrain via exploration strategy. For approach, the robot is equipped with a compass to detect the direction orienting the robot to its goal point. In unknown terrain, only the visible parts of obstacles are represented by the sensory readings. A local obstacle avoidance system is used to avoid collision when obstacles are encountered. The robot records the coordinate of the hit point on which the robot bumps the obstacle when the distance between the robot and the obstacle is below a certain threshold value, which is necessary to be taken into account according to the sensor range. The robot gathers map information about its vicinity and records these data on a partial map. After acquiring enough data, the robot generates a partial map of specific regions of the terrain connecting the start point and goal point as a global resemblance of the entire terrain. Since our objective is for the robot to navigate itself to a goal point, this partial map is enough for geodesic planning by the fast marching method (FMM).

2.2 Bug Algorithms

Bug algorithms operate under some assumptions. First, the localization ability of the robot is perfect. Second, the sensors of the robot are ideal. Different Bug algorithms may require different types of sensors or different environment settings, but all Bug algorithms share the same basic property in that they can be divided into two parts: a goal pursuing mode and a boundary following mode. In goal pursuing mode, the robot in general moves directly toward to the goal until the robot hits an obstacle. When it encounters an obstacle, the robot switches from goal pursuing mode to boundary following mode and does not switch back to goal pursuing mode until some switching condition is satisfied. If the robot does not have a partial map generated that connects the start and destination points, the algorithm will recognize this situation, terminate its running process and report that the destination point is unreachable.

Fig. 1 Three single-obstacle test maps of size 200x200 for experiment. The blue circle is the start; the red circle is the goal. (a) the cavity map. (b) the Sankar map. (c) the zigzag map



3 Experiments and Results

The three environments described by the bitmap file shown in Fig. 1 were used for comparative study of path planning performance of Bug1, Bug2, and BFFMM. All the simulations are run with MATLAB on a PC with Intel Core 2 Duo 1.8 GHz microprocessor.

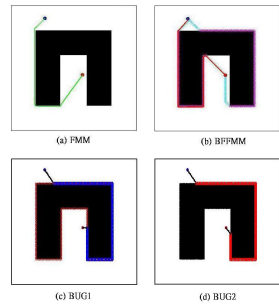


Fig. 2 Simulations on the Cavity map: path generated by (a) FMM (b) BFFMM (c) BUG1 (d) BUG2

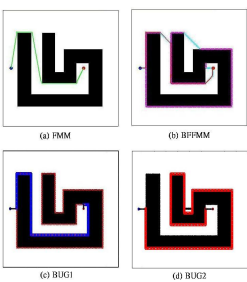


Fig. 3 Simulations on the Sankar map: path generated by (a) FMM (b) BFFMM (c) BUG1 (d) BUG2

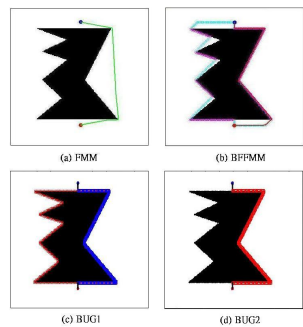


Fig. 4 Simulations on the Zigzag map: path generated by (a) FMM (b) BFFMM (c) BUG1 (d) BUG2

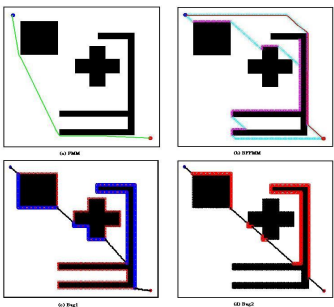


Fig. 5 Simulations on the multiple obstacles map: path generated by (a) FMM (b) BFFMM (c) BUG1 (d) BUG2

Table 1 The length of alternative paths generated by each algorithm in three single-obstacle test maps of size 200x200. The two numbers in the row of BFFMM represent the path length in the two stages of BFFMM respectively. The first number is the path length in exploring phase; the second one is the path length in path-planning phase

Algorithm	Path length on Cavity map	Path length on Sankar map	Path length on Zigzag map
FMM	245.62	261.46	227.74
BFFMM	635.14 / 311.11	858.48 / 279.36	633.07 / 257.19
BUG1	990	1438	903.6
BUG2	337	784	273.64

We conduct simulations with the assumption that the world is perfect, noise-free, and static. The results of the simulations are shown in Fig. 2-4 and summarized in Table 1.

As a reference for evaluation of path length, the path generated by FMM is the geodesic connecting the start and the destination points on the obstructed map, or

the shortest path between the two points. We express the path length as percentage above optimal (PAO), which is defined as

$$\text{PAO} = \left(\frac{\text{generated path length}}{\text{path length generated by FMM}} - 1 \right) \cdot 100$$

to facilitate the comparison between different methods. The comparison is given in Table 2. From the results, the path length generated by Bug1 is the longest because the robot has to contour to the whole boundary curve of the encountered obstacle to determine the switching point. Bug2 generally produces shorter paths than Bug1 because Bug2 stops following the boundary of the obstacle before encircling it. Although the path generated by Bug 2 is shorter than Bug1, Bug2 does not know which side or which route would be better. BFFMM builds the partial map by contouring the boundary curves of obstacles, so the path length in exploring mode would approach the limit of Bug1. The total path length of BFFMM is close to that of Bug1, but the path length of the path planning phase is even shorter than that of Bug2. Besides, the robot could plan a shortest path for a new start-goal pair on the built partial map without resorting to the exploring phase again. Without following the boundary of the obstacles, the path length of FMM is not directly or explicitly influenced by the perimeters of the obstacles. Therefore, this comparison also quantitatively reveals the impact of environment uncertainties on length of paths generated by local navigation algorithms relying on contouring the boundary curve of the obstacle.

3.1 *Effect of Multi-obstacles*

To inspect the effect of multi-obstacles on the algorithms, we perform another simulation shown in Fig. 5 and Table 3. Through observation, we found that as the number of obstacles increased, the difference between the total path lengths generated by BFFMM and Bug1 clearly increased. This phenomenon resulted from the robot operating using Bug1 where it would encircle all the enroute obstacles encountered on the way towards the goal before the switching condition was satisfied. In contrast, when the robot is navigated by BFFMM, it could switch from boundary following mode earlier than when it operated using Bug1 and spent less effort on exploration. Overall, the BFFMM strikes a good balance between less exploration load and shorter path length even though the complexity of the environment increased.

Table 2 PAO of path generated by each algorithm

Algorithm	Cavity map	Sankar map	Zigzag map
BFFMM	159 / 27	228 / 7	1178 / 13
BUG1	303	450	297
BUG2	37	200	20

Table 3 PAO of path generated by each algorithm in multiple obstacles environment

Algorithm	Path length	Relative path length
FMM	306.33	1.00
BFFMM	947.10 / 333.64	3.09 / 1.09
BUG1	1843.53	6.02
BUG2	611.42	2.00

4 Conclusions

Comparisons of path lengths generated by the three local path planning algorithms utilizing boundary following BF techniques are presented. Although paths generated by these kinds of algorithms are far from optimal in path length, they are successful in navigating a mobile robot safely and reliably to a goal along the boundary curve of an obstacle even in situations where the sensor(s) is faulty. From simulations in test environments, we have observed that Bug1 algorithm produces the longest path length, while the BFFMM plans paths that are shorter than both the Bug1 and Bug2. Taking the path length of the exploring phase into account, the BFFMM generates paths with total length approaching those of Bug1 in single obstacle maps and may be even shorter in multiple obstacle maps. Furthermore, new paths could be re-planned by BFFMM easily for a variety of start and destination point pairs within already built partial maps. If the passage ways are known, we can save a lot of computational effort in pursuing destination points with changeable coordinates.

Acknowledgments. The research was supported by National Science Council under contract NSC 96-2221-E-001-018-MY2.

References

1. Chiang, C.H., Liu, J.S.: Boundary Following in Unknown Polygonal Environment Based on Fast Marching Method. In: IEEE International Conference on Advanced Robotics and its Social Impacts (2008)
2. Chiang, C.H., Chiang, P.J., Fei, J.C.C., Liu, J.S.: A Comparative Study of Implementing Fast Marching Method and A* Search for Mobile Robot Path Planning in Grid Environment: Effect of Map Resolution. In: IEEE Workshop on Advanced Robotics and its Social Impacts, pp. 7–12 (2007)
3. Lumelsky, V.J., Stepanov, A.A.: Dynamic Path Planning for a Mobile Automation with Limited Information on the Environment. IEEE Transactions on Automatic Control AC-31(11), 1058–1063 (1986)
4. Ng, J., Braunl, T.: Performance Comparison of Bug Navigation Algorithm. Journal of Intelligent and Robotic Systems 50, 73–84 (2007)
5. Langer, R.A., Coelho, L.S., Oliveira, G.H.C.: K-Bug, A new bug approach of mobile robot path planning. In: 16th IEEE International Conference on Control Application, Part of IEEE Multi-conference on Systems and Control, Singapore, pp. 403–408 (2007)

6. Rao, N.S.V., Shi, S.K.W., Ivengar, S.S.: Robot Navigation in Unknown Terrains: Introductory Survey of Non-Heuristic Algorithms. Technical Report, Oak Ridge National Laboratory (1999)
7. Sethian, J.A.: Level Set Methods. In: Evolving interfaces in geometry, fluid mechanics, computer vision, and materials science. Cambridge University Press, Cambridge (1999)
8. Kamon, I., Rivlin, E., Rimmon, E.: A New Range-Sensor Based Globally Convergent Navigation Algorithm for Mobile Robots. In: Proceedings for the 1996 IEEE International Conference on Robotics and Automation, Minneapolis, Minnesota (1996)
9. LaValle, S.M.: Planning Algorithms. Cambridge University Press, Cambridge (2006)